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TAPLIDS: A Temporal-Aware Prototype-based Lightweight and Interpretable Intrusion Detection System for Intelligent Transportation Systems

Danish Javeed ^a,^{*,1}, Muhammad Adil ^b,¹, Shifa Shoukat ^c, Muhammad Shahid Saeed ^d

^a College of Artificial Intelligence, Dalian Maritime University, Dalian 116026, China

^b College of Intelligence and Computing, Tianjin University, Tianjin, China

^c Software College, Northeastern University, Shenyang 110169, China

^d Dalian University of Technology, Dalian, China

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ABSTRACT

The rapid deployment of Intelligent Transportation Systems (ITS) has revolutionized traffic management and vehicular safety, but it has also introduced significant cybersecurity vulnerabilities. ITS networks consist of heterogeneous sensors, vehicular telemetry, and communication infrastructure, which are increasingly targeted by sophisticated and evolving cyber-attacks, including stealthy, low-frequency, and distributed attacks. Traditional Intrusion Detection Systems (IDSs) are often insufficient in this context, i.e., they lack temporal modeling of dynamic data, fail to handle heterogeneous features effectively, use static classifiers, and provide limited or no interpretability, which undermines their practical applicability in real-world ITS deployments. To address the aforementioned challenges, we propose “TAPLIDS”, a Temporal-Aware Prototype-based Lightweight and Interpretable security framework for ITS environments. TAPLIDS captures temporal dependencies using a sliding-window feature extraction mechanism and ensures consistent feature scaling through normalization. Further, it employs an attention-based embedding module to emphasize the most informative features. Residual latent representations preserve essential original information, while a multi-prototype classification strategy models the diversity of attack behaviors. TAPLIDS also offers interpretable predictions via attention weights, prototype distances, and post-hoc explanations generated using Local Interpretable Model-agnostic Explanations (LIME), enabling actionable insights for security operators and greater trust in its predictions. Extensive experiments demonstrate the effectiveness, efficiency, and interpretability of TAPLIDS. On the CICIoV2024 dataset, TAPLIDS achieves 98.60% accuracy and a 94.43% macro F1-score, outperforming the nearest baseline by 12.91% and 10.46%, respectively. On the N-BaIoT dataset, it attains 96.45% accuracy and a 92.67% macro F1-score, with gains of 8.92% and 13.15% over the nearest baseline. Furthermore, TAPLIDS maintains a lightweight footprint with 11.88 KB and 52 KB model sizes and inference latencies of 0.05 ms and 0.043 ms on the respective datasets, demonstrating its suitability for real-time, resource-constrained ITS environments. Collectively, these results establish TAPLIDS as an accurate, lightweight, interpretable, and readily deployable IDS solution for modern ITS environments.

1. Introduction

The rapid growth of Intelligent Transportation Systems (ITS) has fundamentally transformed modern transportation (Khalil et al., 2024). It enables real-time monitoring of vehicles, traffic flows, and roadside infrastructure, and supports applications such as adaptive traffic control, autonomous vehicle navigation, and smart city planning (Mohanty et al., 2025; Creß et al., 2023; Dui et al., 2024). ITS networks consist of a heterogeneous mix of components, including vehicular sensors (GPS,

accelerometers, cameras, LiDAR) (I. Ahmad et al., 2025), Roadside Units (RSUs) (Nembou Kouonchie et al., 2025), traffic control devices, and wireless communication links such as Dedicated Short Range Communications (DSRC) and cellular V2X (Vehicle-to-everything) (Hussein et al., 2022). This complex integration generates large volumes of temporally correlated, high-dimensional, and multi-modal data, which are crucial for system optimization but also introduce multiple cybersecurity vulnerabilities (Adil et al., 2026). ITS networks are increasingly

* Corresponding author.

E-mail addresses: Danish.J@ieee.org (D. Javeed), 6122000014@tju.edu.cn (M. Adil), 2327006@stu.neu.edu.cn (S. Shoukat), shahidsaeedrana@gmail.com (M.S. Saeed).

¹ Danish Javeed and Muhammad Adil contributed equally to this work.

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exposed to advanced cyber-attacks (Abdo et al., 2023). Some of the common attacks in these networks are Distributed Denial-of-Service (DDoS) attacks targeting network bandwidth (Usha et al., 2025), malware injections that compromise vehicular controllers (Liu et al., 2025), spoofing attacks that manipulate sensor readings or GPS signals (Mohammadi et al., 2025), and low-frequency or stealthy attacks that subtly alter system behavior over time (Youssef et al., 2025). Such threats can not only disrupt traffic operations but also compromise passenger safety and critical infrastructure, highlighting the high stakes of ITS security (Usama et al., 2024).

Existing solutions for intrusion detection in ITS face several critical challenges. Many traditional Intrusion Detection Systems (IDS) methods operate on static snapshots of system data. This prevents them from capturing temporal dependencies and evolving attack patterns that are characteristic of ITS networks. Furthermore, ITS datasets are inherently heterogeneous. It combines multi-modal vehicular, sensor, and network information with varying scales, distributions, and noise characteristics (Bari et al., 2025; Yan and Li, 2025). This heterogeneity across datasets not only complicates feature extraction and modeling but also increases the computational complexity of IDS. Additionally, existing IDS approaches rely on complex DL models or Ensemble Learning (EL) techniques that require extensive training and inference resources (Muneer et al., 2024). Moreover, many IDS frameworks employ single-prototype or shallow classification models (EL Asry et al., 2024), which limit their capacity to capture the diverse and dynamic nature of attacks across different ITS components. Consequently, it leads to higher misclassification rates (Abdulganiyu et al., 2023). Finally, a persistent limitation of existing IDS approaches is the lack of interpretability. Security operators are often unable to understand why a detection alert was triggered. This lack of interpretability results in a reduction in trust and delays the timely mitigation of cyber threats in safety-critical scenarios (Akshya et al., 2025; Nwakanma et al., 2023).

To this end, there is a pressing need for lightweight yet robust and interpretable IDS for ITS, which is capable of handling temporal dependencies, heterogeneous data of ITS networks, and diverse behaviors of advanced cyber threats without imposing prohibitive computational costs.

1.1. Contributions

To address the limitations of existing IDS approaches for ITS environments, we propose TAPLIDS, a Temporal-Aware Prototype-based Lightweight and Interpretable IDS framework designed for efficient and explainable security analytics in dynamic ITS environments. Unlike existing approaches that typically address temporal modeling, feature representation, prototype-based classification, and explainability independently, TAPLIDS integrates these capabilities into a unified lightweight framework. The main contributions of this work are summarized as follows:

- We design a sliding-window temporal feature extraction framework that explicitly models evolving, intermittent, and low-frequency attack behaviors in ITS data streams, enabling robust detection of stealthy and time-dependent attacks in real-time operational settings.
- We develop an attention-based multimodal embedding mechanism that effectively handles the heterogeneity and high dimensionality of ITS datasets by adaptively learning discriminative feature representations across heterogeneous sources, including vehicular sensor readings and network telemetry.
- We propose a residual latent representation learning strategy integrated with a multi-prototype classification mechanism, where each class is represented by multiple learned prototypes. This formulation captures intra-class variability, improves class separation in latent space, and enhances robustness against diverse and evolving attack patterns.

- We introduce an interpretable IDS decision framework that leverages attention weights, prototype distance reasoning, and post-hoc LIME explanations, enabling transparent and trustworthy predictions suitable for deployment in safety-critical ITS environments.

The remainder of this article is organized as follows. Section 2 discusses the related work. Section 3 presents the proposed TAPLIDS framework in detail. Section 4 describes the experimental setup, while Section 5 presents the performance evaluation of the proposed framework, explainability analysis, and comparisons with existing IDS frameworks. Finally, Section 6 concludes the article and outlines directions for future work.

2. Related work

Developing DL-based IDS for ITS is a research hotspot. The researchers in Wazid et al. (2025) have proposed a DL-based malware attack detection for IoT-enabled ITS, namely “DLMA-IITS”. The proposed framework contains an ensemble learning approach by employing a Convolutional Neural Network (CNN) with Gated Recurrent Unit (GRU) and Long Short-Term Memory (LSTM). The researchers in Yanmin et al. (2025) have proposed another IDS for the Internet of Vehicles (IoV) by incorporating the neural coordination between LSTM and GRU. The model is capable of capturing long-term and short-term dependencies among the cyber attack patterns. The authors of Bhavsar et al. (2024) employed Logistic Regression (LR) and CNN in an FL setting to develop an intelligent IDS for transportation IoT networks. The proposed approach is evaluated using NSL-KDD and the Car-Hacking dataset in terms of accuracy and losses. The researchers in Barve and Patheja (2024) have proposed a hybrid IDS by combining CNN and Bidirectional LSTM for Vehicular Ad Hoc Networks (VANET). The authors have simulated a clustering mechanism among the VANET traffic using the K-means clustering algorithm, and SMOTE is adopted to balance data imbalances for efficient processing. The proposed approach has shown reliable performance when evaluated on the VDDD dataset. Similarly, the researchers in Suman et al. (2024) have proposed another DL-based attack detection approach for VANET by employing CNN and Adaptive Neuro Fuzzy Inference Systems (ANFIS). The framework mainly focuses on botnet, sybil, wormhole, portscan, and blackhole attacks, etc. The performance of the proposed approach is examined using i-VANET, ToN-IoT, and CICIDS2017 datasets, where evaluation results validate the efficiency of the framework.

Furthermore, the authors in Khan et al. (2025) proposed a GenAI-based framework for ITS. They focused on mitigating AI-driven attacks using intrusion detection, real-time monitoring, and adaptive security mechanisms. The authors achieved strong performance in detection accuracy and privacy protection compared to existing methods. The authors of Abbass et al. (2025) proposed a cross-layer federated and trust-aware framework for securing V2X communication in adversarial environments. They integrate multi-modal anomaly detection, delay-aware trust estimation, and FL for distributed security without central aggregation. Their proposed IDS also employs multi-objective optimization to balance detection accuracy, collision risk, and communication overhead. Authors in Ahmed and Hossein Anisi (2025) proposed a framework by integrating an IDS with a chaotic-map authenticated key agreement protocol to secure autonomous vehicle communications. Their proposed IDS also leverages Physical Unclonable Functions (PUFs) for secure key generation and supports resilient communication among vehicles, charging stations, and service providers. Researchers in Hellemans et al. (2025) proposed an efficient IDS for in-vehicle networks, namely “MR-TCN”. Their proposed IDS uses a modular reduced temporal convolutional architecture optimized for hardware acceleration and evaluates performance across multiple IVN protocols, including CAN CC, CAN FD, and Automotive Ethernet. The authors in I. Ahmad et al. (2025) proposed a stacking ensemble-based IDS for connected and autonomous vehicles. They combined multiple ML models,

i.e., RF, SVM, AdaBoost, and XGBoost, with an LSTM-RNN meta-learner to capture temporal traffic patterns. Their proposed framework further integrates model-agnostic meta-learning to improve adaptability to unseen and evolving cyberattacks. The authors in Heidari et al. (2026b) proposed FedIoV, a secure and adaptive FL-based IDS for real-time IoV environments. FedIoV combines secure communication, a two-stage aggregation mechanism using TOPSIS and Multi-Krum, a lightweight KANConvNet classifier, and genetic algorithm-based hyperparameter optimization to enhance intrusion detection, privacy, and resource efficiency in heterogeneous vehicular networks. Similarly, the authors of Heidari et al. (2026a) proposed VITA, a lightweight and self-supervised IDS for IoV environments. VITA integrates an adversarial autoencoder with efficient channel attention and residual Temporal Convolutional Networks (TCNs) to capture important spatial features and long-range temporal dependencies without requiring labeled data. The framework demonstrates improved detection accuracy, reduced latency and false positive rates, and lower memory and computational requirements, making it suitable for real-time edge-based IoV deployments. Further, the authors in Heidari and Al-Karaki (2026) proposed NOVA, a self-supervised graph-based framework for detecting behavioral and structural anomalies in IoV networks without labeled data. NOVA models vehicular communications as attributed graphs and combines intimacy-guided subgraph sampling with a Graph Convolutional Network (GCN)-based generative module and a contrastive learning module to identify anomalous behavior and structural inconsistencies. The framework achieves high detection accuracy, real-time throughput, and near-linear scalability for large-scale IoV environments.

Moreover, Prototype-based learning has been explored in IDSs to improve generalization under limited labeled data. Prototypical Networks represent each class using a prototype in an embedding space and classify samples based on distance to these prototypes. For example, the authors in Suleymanov et al. (2025) proposed a semi-supervised Prototypical Network (SSPNet) for intrusion detection. They enhance standard prototypical networks by leveraging pseudo-labels from high-confidence samples to improve representation learning and generalization to unseen attack classes. The approach demonstrates improved performance in few-shot intrusion detection scenarios compared to conventional prototypical networks. Similarly, the authors in Kinger and Rupa (2025) proposed a hybrid intrusion detection framework combining Prototypical Network-based meta-learning with Mahalanobis distance-based outlier detection. The method integrates few-shot learning with statistical anomaly detection to improve the classification of known attacks and enable the detection of unseen or zero-day attacks, demonstrating strong adaptability in evolving cybersecurity environments. Finally, the authors in J. Ahmad et al. (2025) proposed a stochastic Prototypical Network based on random neural networks for few-shot intrusion detection in CAN networks. The method models uncertainty through stochastic prototypes, improving robustness to noisy traffic and enhancing detection reliability while maintaining high accuracy and low inference latency suitable for real-time deployment.

2.1. Research gaps

Although significant progress has been made in intrusion detection for ITS, existing studies remain fragmented across different research directions, focusing individually on accuracy improvement, lightweight design, or explainability. As a result, there is still a lack of unified approaches that jointly address temporal dependency modeling, heterogeneous multimodal representation learning, robust classification, and interpretability within a single framework. In addition, most existing methods either rely on deep models with high computational cost or lightweight architectures with limited representation capability, making them less suitable for real-time ITS deployment. Similarly, prototype-based reasoning and intrinsic interpretability are rarely integrated into end-to-end IDS designs, limiting their ability to provide both accurate and explainable decisions. We address these gaps through a

unified TAPLIDS framework that integrates temporal-aware feature extraction, attention-based multimodal embedding, residual latent representation learning with multi-prototype classification, and interpretable decision-making mechanisms.

3. Proposed temporal-aware prototype-based lightweight and interpretable intrusion detection system (TAPLIDS)

3.1. Problem statement

ITS integrates vehicular sensors, communication networks, and traffic management systems to enhance safety and efficiency. However, the extreme connectivity of ITS devices and sensors exposes them to a wide range of cyber-attacks. These attacks include Distributed Denial-of-Service (DDoS), spoofing, routing manipulation, and sensor data tampering. Traditional IDSs, such as signature-based or conventional anomaly detection methods, often fail in such environments. This is due to the high-dimensional nature of ITS data, the temporal correlations among events, the heterogeneity of sensor sources, and the need for interpretable decisions for auditing and safety assurance.

3.2. Proposed TAPLIDS

We propose a novel Temporal Aware Prototype-based Lightweight and Interpretable Intrusion Detection System (TAPLIDS) for ITSs. We design this IDS to capture temporal patterns in the ITS network traffic and emphasize important features. Further, the proposed IDS preserves essential information in latent embeddings and provides interpretable classifications. TAPLIDS comprises four key modules: (i) *temporal feature extraction*, (ii) *attention-based feature embedding*, (iii) *residual latent representation*, and (iv) *multi-prototype classification*. Each module of the proposed IDS is designed with a clear rationale for addressing specific challenges in ITS cybersecurity. The modular design of the proposed TAPLIDS ensures temporal dynamics are captured, critical features are emphasized, information is preserved in latent space, and decisions are interpretable for audit and analysis purposes. The structural workflow of the proposed TAPLIDS is illustrated in Fig. 1, while Algorithm 1 provides the step-by-step procedure. The details about each module of the proposed TAPLIDS are as follows.

3.2.1. Temporal feature extraction

Consider the ITS dataset represented as:

$$\mathcal{X} \in \mathbb{R}^{\mathcal{N} \times \mathcal{F}}, \quad (1)$$

where $\mathcal{N}$ denotes the total number of temporal observations and $\mathcal{F}$ denotes the number of features recorded at each time step. To capture short-term temporal dependencies and evolving patterns in the ITS data, we employ a sliding-window strategy with a window length of $\mathcal{W} = 3$ and a stride of one sample. Thus, each window contains three consecutive observations, and the window is shifted by one observation to generate the subsequent temporal instance. This configuration results in an overlap of two observations between consecutive windows. For a dataset containing $\mathcal{N}$ observations, the procedure generates $\mathcal{N} - \mathcal{W} + 1$ temporal windows without padding.

For each feature within a temporal window, three complementary temporal statistics are extracted: (i) *mean*, (ii) *standard deviation*, and (iii) *delta*. These statistics capture the average behavior, temporal variability, and net directional change of each feature, respectively. Specifically, the mean of feature j within window i is calculated as:

$$\mathcal{M} * j^{(i)} = \frac{1}{\mathcal{W}} \sum_{t=i}^{i+\mathcal{W}-1} \mathcal{X}_{t,j}, \quad (2)$$

where $\mathcal{M}_j^{(i)}$ represents the average value of feature j within the window.

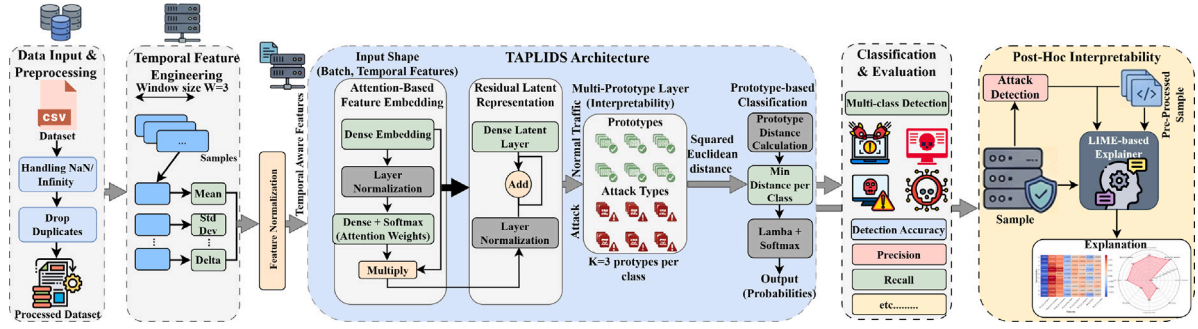


Fig. 1. Proposed TAPLIDS's workflow and architecture.

The standard deviation is calculated as:

$$S * j^{(i)} = \sqrt{\frac{1}{W} \sum_{t=i+W-1}^{i} (\mathcal{X}_{t,j} - \mathcal{M}_j^{(i)})^2}, \quad (3)$$

where $S_j^{(i)}$ captures the temporal variability of feature j within the window.

The delta feature is defined as:

$$\Delta_j^{(i)} = \mathcal{X}_{i+W-1,j} - \mathcal{X}_{i,j}, \quad (4)$$

where $\Delta_j^{(i)}$ represents the net change in feature j across the window and captures its short-term directional behavior.

The three statistics are then concatenated across all F features to construct the temporal representation for window i :

$$\mathcal{Z}^{(i)} = [\mathcal{M}_1^{(i)}, \dots, \mathcal{M}_F^{(i)}, S_1^{(i)}, \dots, S_F^{(i)}, \Delta_1^{(i)}, \dots, \Delta_F^{(i)}]. \quad (5)$$

Consequently, each temporal window produces a $3F$ -dimensional representation that summarizes the statistical characteristics and short-term temporal trends of the observed ITS features. The resulting temporal representations are subsequently used as input to the downstream normalization and attention-based embedding modules.

3.2.2. Feature normalization

The extracted temporal features can have heterogeneous ranges and units, which may negatively affect model convergence and performance. We apply z-score normalization to ensure that all features are on a comparable scale. The mean and standard deviation are computed only from the training dataset, and the same transformation is applied to both training and testing data to avoid data leakage. This is implemented using the StandardScaler function in scikit-learn.

$$\tilde{\mathcal{F}}_j^{(i)} = \frac{\mathcal{F}_j^{(i)} - \mu_j}{\sigma_j} \quad (6)$$

where $\tilde{\mathcal{F}}_j^{(i)}$ is the normalized feature value, μ_j and σ_j are the mean and standard deviation of feature j , computed over the training set.

3.2.3. Attention-based feature embedding

Furthermore, we apply an attention mechanism to capture the most informative aspects of the temporal features. This mechanism allows the model to assign different importance to the dimensions of the learned embedding according to their relevance, emphasizing informative representations while suppressing less informative ones.

First, the input features $\mathcal{X}$ are transformed into a dense embedding:

$$\mathcal{H} = \text{ReLU}(\mathcal{W}_e \mathcal{X} + \mathcal{B}_e) \quad (7)$$

Where $\mathcal{H} \in \mathbb{R}^d$ is the dense feature embedding, $\mathcal{W}_e \in \mathbb{R}^{d \times d}$ is a trainable weight matrix, $\mathcal{B}_e \in \mathbb{R}^d$ is a bias vector, and $\text{ReLU}(\cdot)$ introduces non-linearity to capture complex interactions among features.

Next, attention weights are computed to quantify the relative importance of each feature in the embedding:

$$\alpha = \text{softmax}(\mathcal{W}_a \mathcal{H} + \mathcal{B}_a) \quad (8)$$

where $\alpha \in \mathbb{R}^d$ contains the normalized attention weights assigned to the dimensions of the learned embedding.

Finally, the attention-weighted feature representation is obtained by performing element-wise multiplication between the embedding and attention weights:

$$\mathcal{X}_{att} = \mathcal{H} \odot \alpha \quad (9)$$

where $\mathcal{X}_{att} \in \mathbb{R}^d$ is the attention-enhanced feature vector, and $\odot$ denotes element-wise multiplication.

This representation emphasizes informative latent dimensions while attenuating less relevant ones. Consequently, the prototype-based classifier receives a more discriminative representation, enabling improved separation between benign and malicious traffic patterns and facilitating the identification of diverse attack behaviors.

3.2.4. Residual latent representation

We further employ a residual connection over the attention-weighted features to enable richer non-linear transformations while preserving the original feature information. This approach helps mitigate the problem of vanishing gradients and allows the model to learn incremental refinements over the input representation.

The residual latent embedding is computed as:

$$\mathcal{Z} = \text{ReLU}(\mathcal{W}_l \mathcal{X}_{att} + \mathcal{B}_l) + \mathcal{X}_{att} \quad (10)$$

where $\mathcal{Z} \in \mathbb{R}^d$ is the residual latent embedding, and $\text{ReLU}(\cdot)$ introduces non-linearity. The addition of $\mathcal{X}_{att}$ forms the residual connection, allowing the model to preserve original feature information while learning new patterns.

To stabilize training and further enhance representational quality, the residual embedding is normalized using layer normalization:

$$\mathcal{Z}_{norm} = \text{LayerNorm}(\mathcal{Z}) \quad (11)$$

where $\mathcal{Z}_{norm} \in \mathbb{R}^d$ is the layer-normalized residual embedding. Layer normalization reduces covariate shift across features, accelerates convergence, and improves model generalization.

3.2.5. Multi-prototype classification

To capture intra-class variability and improve classification robustness, each class c is represented by multiple prototypes $\mathcal{P}_{c,k}$, where $k = 1, \dots, K$ denotes the k th prototype of class c . The squared Euclidean distance between the residual latent embedding $\mathcal{Z}$ and each prototype is computed as:

$$\mathcal{D}_{c,k}(\mathcal{Z}) = \|\mathcal{Z} - \mathcal{P}_{c,k}\|_2^2 \quad (12)$$

where $\mathcal{D}_{c,k}(\mathcal{Z})$ is the squared Euclidean distance between $\mathcal{Z}$ and the k th prototype of class c , and $\mathcal{P}_{c,k} \in \mathbb{R}^d$ is the corresponding prototype vector.

Algorithm 1: Proposed TAPLIDS

Input: ITS dataset $\mathcal{X} \in \mathbb{R}^{N \times F}$, labels $\mathcal{Y}$, window size W , prototypes per class K , epochs E

Output: Trained TAPLIDS model with embeddings, attention, and prototypes

- 1 **Temporal Feature Extraction:**
- 2 **for** $i = 1$ **to** $N - W + 1$ **do**
- 3 **for** $j = 1$ **to** F **do**
- 4 Compute mean: $\mathcal{M}_j^{(i)} = \frac{1}{W} \sum_{t=i}^{i+W-1} \mathcal{X}_{t,j}$
- 5 Compute std: $S_j^{(i)} = \sqrt{\frac{1}{W} \sum_{t=i}^{i+W-1} (\mathcal{X}_{t,j} - \mathcal{M}_j^{(i)})^2}$
- 6 Compute delta: $\Delta_j^{(i)} = \mathcal{X}_{i+W-1,j} - \mathcal{X}_{i,j}$
- 7 **end**
- 8 Concatenate features: $\mathcal{Z}^{(i)} = [\mathcal{M}^{(i)}, S^{(i)}, \Delta^{(i)}]$
- 9 **end**
- 10 **Feature Normalization:**
- 11 Compute z-score normalization using training set statistics:

$$\tilde{F}_j^{(i)} = \frac{F_j^{(i)} - \mu_j}{\sigma_j}$$

- 12 **Attention Embedding:**
- 13 Compute dense embedding $\mathcal{H} = \text{ReLU}(\mathcal{W}_e \tilde{F} + B_e)$
- 14 Compute attention weights $\alpha = \text{softmax}(\mathcal{W}_a \mathcal{H} + B_a)$
- 15 Apply attention: $\mathcal{X}_{att} = \mathcal{H} \odot \alpha$
- 16 **Residual Latent Representation:**
- 17 $\mathcal{Z} = \text{ReLU}(\mathcal{W}_l \mathcal{X}_{att} + B_l) + \mathcal{X}_{att}$
- 18 $\mathcal{Z}_{norm} = \text{LayerNorm}(\mathcal{Z})$
- 19 **Multi-Prototype Classification:**
- 20 **for** $c = 1$ **to** C **do**
- 21 **for** $k = 1$ **to** K **do**
- 22 Compute distance $d_{c,k}(\mathcal{Z}) = \|\mathcal{Z} - \mathcal{P}_{c,k}\|_2^2$
- 23 **end**
- 24 $\mathcal{D}_c^{min} = \min_k d_{c,k}$
- 25 **end**
- 26 Compute class probabilities:

$$\mathcal{P}(y = c | \mathcal{Z}) = \frac{\exp(-\mathcal{D}_c^{min}(\mathcal{Z}))}{\sum_{j=1}^C \exp(-\mathcal{D}_j^{min}(\mathcal{Z}))}$$

- 27 **Model Training:**
- 28 All parameters (embedding layers, attention module, and prototypes) are jointly optimized using cross-entropy loss:

$$\mathcal{L}_{CE} = - \sum_{i=1}^N \sum_{c=1}^C \mathcal{Y}_{i,c} \log(\mathcal{P}(y_i = c | \mathcal{Z}_i))$$

Using backpropagation with the Adam optimizer over E epochs.

- 29 **return** Trained TAPLIDS model

For each class, the minimum distance across its prototypes is used to represent the proximity of the embedding to that class:

$$\mathcal{D}_c^{min}(\mathcal{Z}) = \min_{k=1, \dots, K} \mathcal{D}_{c,k}(\mathcal{Z}) \quad (13)$$

where $\mathcal{D}_c^{min}(\mathcal{Z})$ is the closest match between the embedding $\mathcal{Z}$ and any of the prototypes of class c .

Finally, the class probabilities are computed using a softmax over the negative minimum distances, producing a normalized probability distribution over all C classes:

$$\mathcal{P}(y = c | \mathcal{Z}) = \frac{\exp(-\mathcal{D}_c^{min}(\mathcal{Z}))}{\sum_{j=1}^C \exp(-\mathcal{D}_j^{min}(\mathcal{Z}))} \quad (14)$$

where $\mathcal{P}(y = c | \mathcal{Z})$ is the predicted probability that embedding $\mathcal{Z}$ belongs to class c , C is the total number of classes, and the softmax ensures that the probabilities sum to one.

By employing multiple prototypes per class, the model can effectively represent complex class distributions and better handle variations within the same class, improving overall classification accuracy and robustness.

3.2.6. Model training

The proposed model is trained to minimize the discrepancy between the predicted class probabilities and the true labels using the Categorical Cross-Entropy (CC-E) loss:

$$\mathcal{L}_{CE} = - \sum_{i=1}^N \sum_{c=1}^C \mathcal{Y}_{i,c} \log(\mathcal{P}(y_i = c | \mathcal{Z}_i)) \quad (15)$$

where $\mathcal{Y}_{i,c}$ denotes the one-hot encoded ground-truth label of sample i for class c , $\mathcal{P}(y_i = c | \mathcal{Z}_i)$ represents the predicted probability that sample i belongs to class c given its latent representation $\mathcal{Z}_i$, N denotes the total number of training samples, and C denotes the total number of classes.

The loss function encourages the model to assign high probabilities to the correct classes while penalizing incorrect predictions.

3.3. Interpretability

The proposed TAPLIDS model provides interpretability at multiple levels, allowing insights into both feature contributions and classification decisions.

3.3.1. Attention-based interpretability

The attention mechanism produces a weight vector α for each feature:

$$\alpha = \text{softmax}(\mathcal{W}_a \mathcal{H} + B_a) \quad (16)$$

The contribution of each feature j to the final representation $\mathcal{X}_{att}$ can be expressed as:

$$\mathcal{X}_{att,j} = \alpha_j \cdot H_j \quad (17)$$

where α_j is the attention weight for feature j , and H_j is the corresponding dense embedding. Features with higher α_j have a stronger influence on the prediction.

3.3.2. Residual embedding transparency

The residual latent embedding preserves original feature information while allowing non-linear transformation:

$$\mathcal{Z} = \text{ReLU}(\mathcal{W}_l \mathcal{X}_{att} + B_l) + \mathcal{X}_{att} \quad (18)$$

Layer normalization ensures stability:

$$\mathcal{Z}_{norm} = \frac{\mathcal{Z} - \mu_{\mathcal{Z}}}{\sigma_{\mathcal{Z}}} \quad (19)$$

where $\mu_{\mathcal{Z}}$ and $\sigma_{\mathcal{Z}}$ are the mean and standard deviation of the residual embedding across features.

3.3.3. Prototype-based confidence

For each class c with K prototypes $\mathcal{P}_{c,k}$, the minimum distance determines the most representative prototype:

$$\mathcal{D}_c^{min}(\mathcal{Z}) = \min_{k=1, \dots, K} \|\mathcal{Z}_{norm} - \mathcal{P}_{c,k}\|_2^2 \quad (20)$$

where $\mathcal{Z}_{norm}$ represents the normalized latent embedding of the input sample, $\mathcal{P}_{c,k}$ denotes the k th prototype of class c , and $\|\cdot\|_2^2$ represents the squared Euclidean distance. The term $\mathcal{D}_c^{min}(\mathcal{Z})$ indicates the distance to the most representative prototype for class c .

The predicted class probability is computed by converting prototype distances into a probability distribution using a softmax function:

$$P(y = c | \mathcal{Z}) = \frac{\exp(-D_c^{\min}(\mathcal{Z}))}{\sum_{j=1}^C \exp(-D_j^{\min}(\mathcal{Z}))} \quad (21)$$

where $P(y = c | \mathcal{Z})$ denotes the probability that the input embedding $\mathcal{Z}$ belongs to class c , C is the total number of classes, and $D_c^{\min}(\mathcal{Z})$ represents the minimum squared Euclidean distance between $\mathcal{Z}$ and the prototypes of class c . The exponential function converts distances into similarity scores, and the denominator normalizes these scores across all classes to form a valid probability distribution. A smaller distance to a prototype indicates higher confidence in the prediction.

3.3.4. Post-hoc interpretability with LIME

LIME approximates the complex model locally with a linear surrogate model g around an input $\mathcal{Z}_0$:

$$g(\mathcal{Z}') = \beta_0 + \sum_{j=1}^d \beta_j \mathcal{Z}'_j, \quad \mathcal{Z}' \in \mathcal{N}(\mathcal{Z}_0) \quad (22)$$

where β_j represents the local importance of feature j , and $\mathcal{N}(\mathcal{Z}_0)$ denotes the neighborhood of the instance $\mathcal{Z}_0$. The magnitude of β_j indicates the contribution of feature j to the prediction of $\mathcal{Z}_0$. LIME provides local explanations even for highly non-linear or attention-weighted embeddings.

Overall, by combining attention weights, residual embeddings, prototype distances, and LIME feature attribution, TAPLIDS achieves both global and local interpretability, offering insights into which features and patterns are driving the model's decisions.

4. Experimental setup

4.1. Dataset details

We utilized two publicly available datasets, i.e., CICIoV2024 (Neto et al., 2024) and NBaIoT (Meidan et al., 2018), to evaluate the proposed IDS. These datasets have been utilized in the literature to evaluate AI-based IDSs. The CICIoV comprises six attack classes with a benign class, while NBaIoT, on the other hand, has eight attacks and a normal class. In this study, we have considered all the classes of the CICIoV dataset and eight classes of the N-BaIoT dataset. Fig. 2 provides the class types and their instances, while the complete details about the dataset features are provided by the authors in Neto et al. (2024), Meidan et al. (2018). We preprocessed these datasets by using all the necessary steps, i.e., removing NaN and infinity values. Further, we normalized the features using a standard scaler, while for the output label, we employed one-hot encoding. Finally, we divided the dataset into two parts. For the training part, we used 80 percent of the dataset, while the rest of the 20 percent was used for testing the model. Furthermore, we took 10 percent of the data from the training set as validation data to monitor model generalization during training.

4.2. Hardware and software setup

All the experiments were conducted using an ordinary PC equipped with an Intel Core i7-10710U CPU operating at a speed of 1.61 GHz with 32 GB RAM. Regarding the software, we have performed all the experiments using Python and TensorFlow. We utilized the NumPy and pandas libraries for data pre-processing, while matplotlib and seaborn were used for visualization purposes. Furthermore, for the performance evaluation, we have utilized Scikit-learn. The experiments were performed using the Jupyter IDE of Anaconda Navigator.

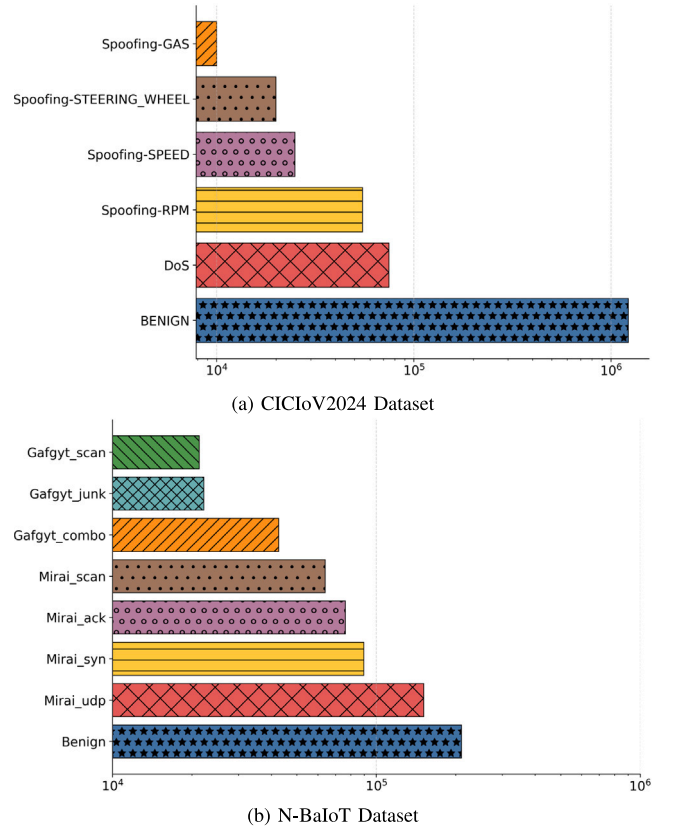


Fig. 2. Dataset classes and number of samples.

Table 1

Hyperparameter Settings of the proposed TAPLIDS.

Component/Layer	Hyperparameter	Value/Description
Input Layer	Input Shape	x_{train} shape
Dense Embedding Layer	Units (d_{model})	32
	Activation	ReLU
	Layer Normalization	Yes
Attention Layer	Units	32
	Activation	Softmax
	Operation	Feature-wise weighting (Multiply)
Residual Latent Layer	Units	32
	Activation	ReLU
	Residual Connection	Yes
	Layer Normalization	Yes
Prototype Layer	Number of Prototypes	$num_classes \times K$
	Prototype Dimension	32
	Trainable	Yes
Distance Computation	Metric	Squared Euclidean Distance
	Reduction	Min per class prototype
Output Layer	Activation	Softmax over negative distances
Training Setup	Optimizer	Adam
	Learning Rate	0.001
	Loss Function	Categorical Crossentropy
	Batch Size	32
	Epochs	20
Model Setting	Prototypes per class (K)	3

4.3. Hyperparameter

The proposed IDS was implemented using the TensorFlow and Keras libraries. Before training, temporal statistical features were extracted using a sliding window approach to capture short-term traffic dynamics in network flows. A temporal window of size three was applied to the input sequence. For each feature within the window, three statistical descriptors were computed: mean, standard deviation, and

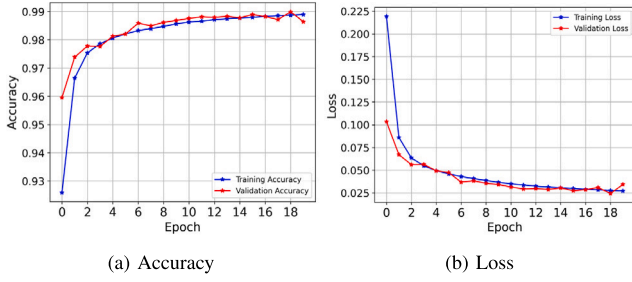


Fig. 3. Training versus validation accuracy and loss analysis for CICIv2024 Dataset.

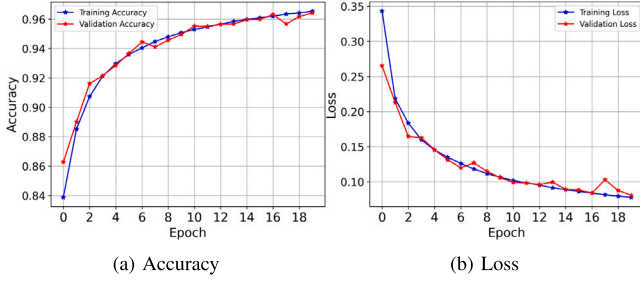


Fig. 4. Training versus validation accuracy and loss analysis for N-BaIoT dataset.

temporal difference (δ). This process enhances temporal awareness while maintaining a lightweight representation suitable for real-time intrusion detection. The processed features were normalized using the StandardScaler to ensure zero mean and unit variance, which stabilizes gradient-based optimization. The model projects input features into a latent embedding space using a dense layer with ReLU activation followed by layer normalization. An attention-based weighting mechanism is then applied to highlight informative features. A residual connection is introduced to improve feature representation and training stability. For classification, the model employs a prototype-based learning mechanism where each class is represented by multiple learnable prototypes in the latent space. During inference, the Euclidean distance between the latent embedding and the prototypes is computed, and the minimum distance per class is used to determine the class probabilities via a softmax function. The model was trained using the Adam optimizer with categorical cross-entropy loss. Training was conducted for 20 epochs with a batch size of 32. The complete hyperparameter setup is provided in Table 1.

5. Performance analysis & comparison

5.1. Training vs. Validation analysis

This section presents performance analysis results of the proposed TAPLIDS framework during the training and validation phase. The accuracy and loss analysis using the CICIv2024 dataset is presented in Fig. 3. Fig. 3(a) refers to the accuracy comparison, whereas Fig. 3(b) represents the comparison of training and validation losses. It can be seen that the TAPLIDS have demonstrated a steady growth in accuracy in the initial few epochs, leading to stable progress in the later epochs. This indicates a stable learning behavior of the proposed model on the training and validation data. Moreover, the near-identical performance between training and validation accuracy further indicates the strong generalization capabilities of TAPLIDS. The training loss and validation loss also show a similar pattern, where a sudden drop in both losses is observed in the initial few epochs. It indicates the effective optimization of the TAPLIDS framework, and a smaller gap between the losses suggests stable convergence. Similar trends have been witnessed

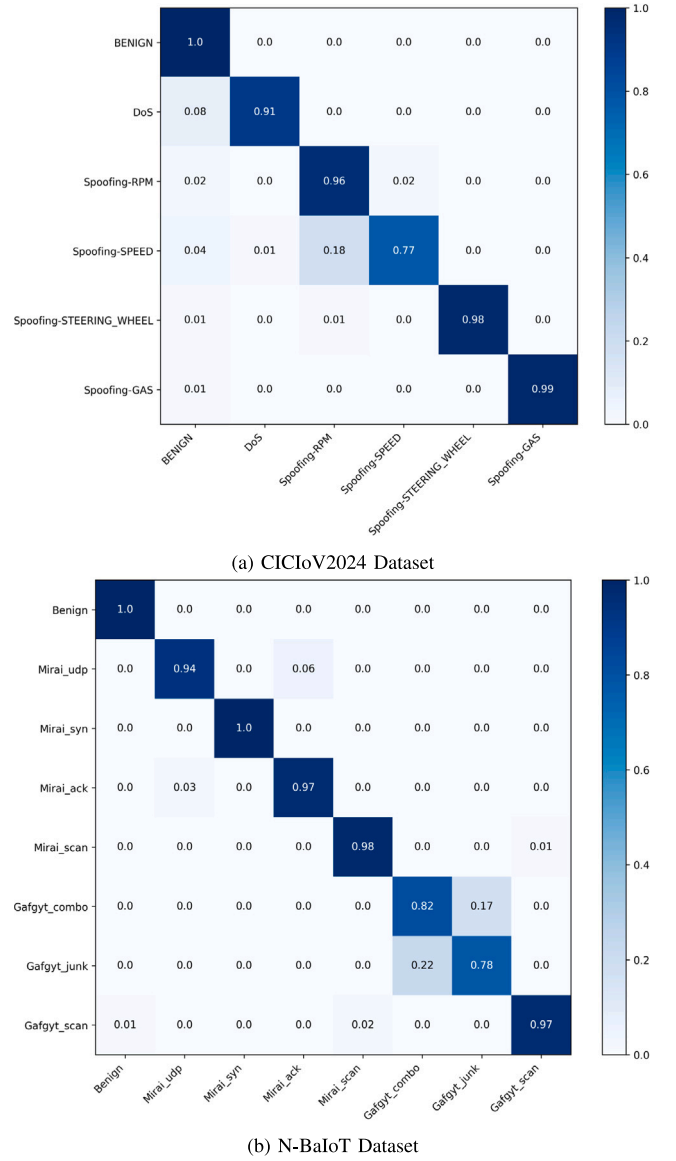


Fig. 5. Confusion matrix analysis.

while evaluating the performance of the proposed TAPLIDS framework on the N-BaIoT dataset, as shown in Fig. 4. The accuracy comparison shown in Fig. 4(a) represents the fast learning characteristics of the TAPLIDS framework as the accuracy curve tends to rapidly increase in the starting epochs. After that, a steady growth behavior has been noticed. Fig. 4(b) points toward the comparison of losses, where both loss curves show a continuous decrease in the following epochs, referring to effective learning and optimization.

5.2. Confusion matrix and ROC curve analysis

The confusion matrix analysis of the proposed TAPLIDS framework is presented in Fig. 5, where Fig. 5(a) reflects the performance on CICIv2024, whereas Fig. 5(b) depicts performance using the N-BaIoT datasets. The strong diagonal values close to 1 indicate that the proposed framework correctly classifies the majority of the samples. In the case of the CICIv2024 dataset, the model demonstrates strong performance across several classes, including Benign, Spoofing-GAS, and Spoofing-STEERING_WHEEL, among others. However, as shown in Fig. 5(a), the Spoofing-SPEED class exhibits a comparatively lower recall of 0.7673, which is the lowest among all classes. This behavior

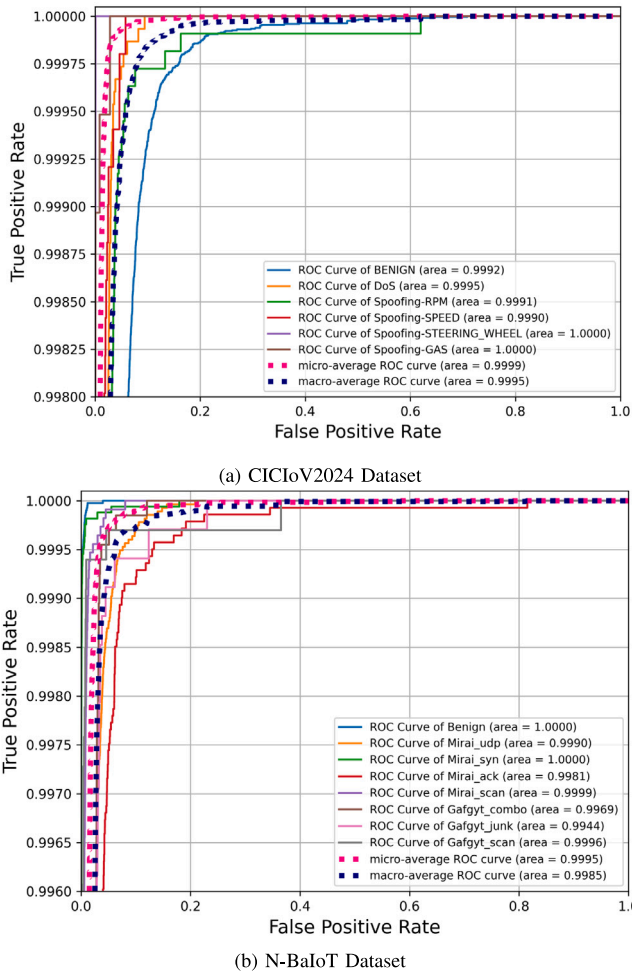


Fig. 6. ROC curve analysis.

is primarily attributed to two factors: (i) the significant class imbalance present in the dataset, where Spoofing-SPEED instances are underrepresented, and (ii) the high behavioral similarity between Spoofing-SPEED attacks and normal driving speed fluctuations, which increases ambiguity during classification. Despite this challenge, the model maintains a high precision for this class, i.e., 0.92080, as shown in Table 2, indicating a conservative detection behavior that prioritizes reducing false alarms. Similarly, while classifying the traffic from the N-BaIoT dataset, the TAPLIDS framework has also correctly identified severe attacks, including Mirai-syn, Mirai-scan, Gafgyt-scan, etc. On both datasets, the model has shown very limited misclassifications that occurred between closely overlapping attack patterns. However, the TAPLIDS framework has demonstrated an overall satisfactory performance with reliable and robust multiclass attack detection in intelligent transportation networks. The performance is further illustrated using the ROC curve given in Fig. 6, where Fig. 6(a) and Fig. 6(b) reflect analysis using the CICIoV2024 and N-BaIoT datasets, respectively. The higher ROC performance suggests the efficiency of the proposed model in achieving high true positives while minimizing the number of false positives. It can be seen that most of the curves are very close to the top-left corner in both datasets. The Area Under the Curve (AUC) values are very impressive, corresponding to the strong generalization capabilities of the TAPLIDS framework. The AUC values range from 0.9990 to 0.9999 for CICIoV2024, whereas they range from 0.9944 to 0.9999 for the N-BaIoT dataset. These results clearly exhibit the credibility of the proposed model in separability, classifying among the benign and closely overlapping complex attack classes.

Table 2

TAPLIDS class-wise performance on CICIoV2024 dataset.

Class Name	Precision	Recall	F1-score	FP Rate	AUC-ROC
Benign	0.99313	0.99633	0.99473	0.04564	0.9992
DoS	0.97839	0.91401	0.94510	0.00114	0.9995
Spoofing-RPM	0.87285	0.95690	0.91294	0.00560	0.9991
Spoofing-SPEED	0.92080	0.76734	0.83709	0.00120	0.9990
Spoofing-STEERING_WHEEL	0.98427	0.98327	0.98377	0.00022	0.9999
Spoofing-GAS	0.99124	0.99328	0.99226	0.00006	0.9999

Table 3

TAPLIDS class-wise performance on N-BaIoT dataset.

Class Name	Precision	Recall	F1-score	FP Rate	AUC-ROC
Benign	0.99839	0.99888	0.99864	0.00078	0.9999
Mirai_udp	0.98424	0.94068	0.96197	0.00433	0.9990
Mirai_syn	0.99799	0.99588	0.99693	0.00030	0.9999
Mirai_ack	0.89343	0.97016	0.93022	0.01491	0.9981
Mirai_scan	0.99329	0.98372	0.98848	0.00066	0.9999
Gafgyt_combo	0.87404	0.82225	0.84736	0.00677	0.9969
Gafgyt_junk	0.69343	0.77958	0.73399	0.00974	0.9944
Gafgyt_scan	0.94150	0.97250	0.95675	0.00166	0.9996

5.3. Class-wise performance analysis

The performance of the proposed TAPLIDS framework is thoroughly examined under all the performance metrics across all major classes of both datasets. The performance analysis using the CICIoV2024 dataset is presented in Table 2. The TAPLIDS framework has performed a perfect classification among all the addressed classes. The model has assured precise detection of the benign class with 0.9931 precision, 0.99633 recall value, and a very minimal false positive rate of 0.045. Meanwhile, similar promising results have been witnessed in the case of attack classes. The model has detected the Spoofing_GAS class with 0.99124 precision and with an extremely low 0.00006 false positive rate. Similarly, the Spoofing-STEERING_WHEEL class is also classified with 0.9824 precision, 0.9837 recall, and a 0.0002 false positive rate. Furthermore, the Spoofing-SPEED class achieves a precision of 0.92080 and a recall of 0.76734, resulting in an F1-score of 0.83709. The comparatively lower recall indicates that this class is more challenging to detect reliably. This behavior is primarily attributed to the high similarity between Spoofing-SPEED attacks and normal driving speed variations, as well as the relatively imbalanced distribution of this class in the dataset, which makes learning discriminative boundaries more difficult. Nevertheless, the overall results in Table 2 demonstrate strong and stable detection performance across spoofing categories. Such results demonstrate the efficiency of TAPLIDS for reliably classifying complex classes. While inspecting the performance using the N-BaIoT dataset in Table 3, the proposed TAPLIDS framework achieves consistently high performance across most classes in the N-BaIoT dataset, particularly for Mirai-based attacks and Gafgyt_scan, where strong precision, recall, and AUC-ROC values are observed. However, a comparatively lower performance is recorded for the Gafgyt_junk class, which achieves an F1-score of 0.73399 and a precision of 0.69343. This reduction in performance can be attributed to the inherently noisy and unstructured nature of junk traffic, which lacks consistent behavioral patterns and exhibits high variability across samples. As a result, Gafgyt_junk overlaps significantly with benign and other attack distributions in the feature space, making it more difficult to learn robust discriminative boundaries. In addition, the weak structural characteristics of this traffic type further increase the likelihood of misclassification.

Despite this challenge, the model maintains strong performance across the majority of attack categories in N-BaIoT, demonstrating its overall robustness and effectiveness in detecting well-structured attack behaviors. The observed limitation for Gafgyt_junk highlights the inherent difficulty of modeling highly heterogeneous and weakly structured network traffic in real-world IoT intrusion detection scenarios.

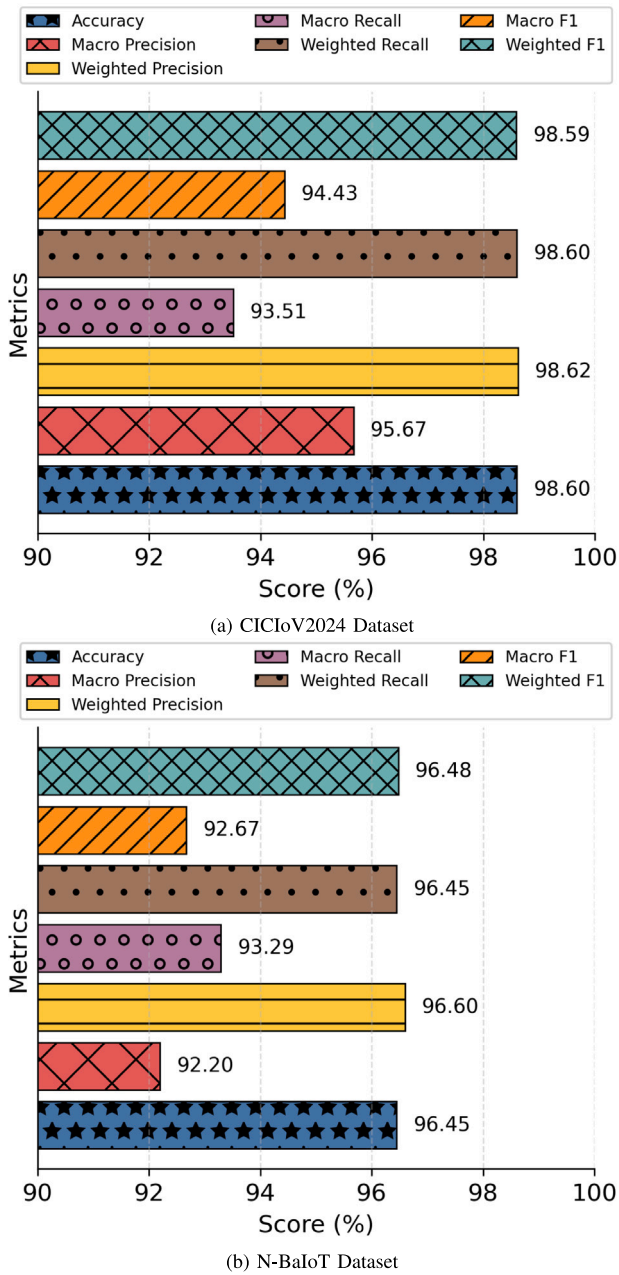


Fig. 7. TAPLIDS overall performance.

Table 4
TAPLIDS computational efficiency.

Metrics	CICIoV2024 Dataset	N-BaIoT Dataset
Parameters	3040	13,312
M-FLOPS	8.014	29.64
Throughput (samples/s)	628.49	1039.17
Inference latency per Sample (ms)	0.05	0.043
Model Size (KB)	11.88 KB	52.00 KB

5.4. Average performance analysis

We provide the overall performance of the proposed TAPLIDS in Fig. 7 regarding detection accuracy, macro and weighted average precision, recall, and F1-score. The proposed IDS has a high detection accuracy of 98.60% for CICIoV2024 and 96.45% for the N-BaIoT dataset. Further, it has macro and weighted precisions of 95.67% and 98.62% for

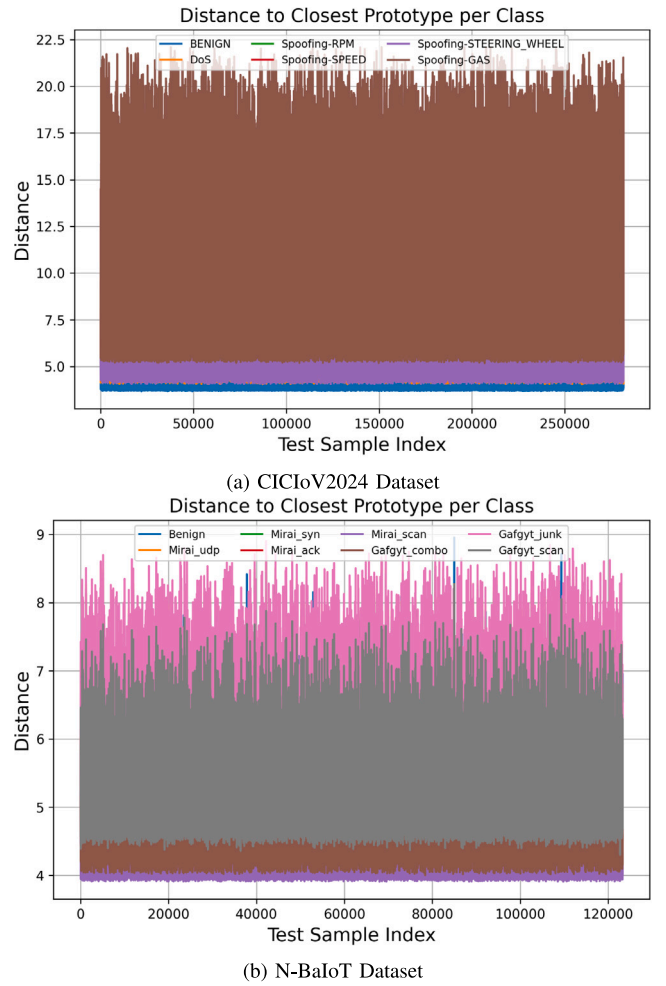


Fig. 8. Distance of test samples to the closest prototype for each class.

CICIoV2024 and 92.20% and 96.60% for N-BaIoT. Regarding recall, the proposed IDS has a macro recall of 93.51% and 93.29% with weighted recall of 98.60% and 96.45%, respectively. Finally, the proposed IDS has also achieved higher F1-score values for both datasets, where it has macro F1-scores of 94.43% and 92.67% and weighted F1-scores of 98.59% and 96.48%.

5.5. Model efficiency

Table 4 provides the computational complexity of the proposed TAPLIDS framework. It can be seen that the proposed IDS has very few parameters for both datasets, i.e., 3040 for CICIoV2024 and 13,312 for the N-BaIoT dataset. Further, it has Mega Flops (M-Flops) of 8.014 and 29.64, respectively. Regarding the throughput, the proposed IDS has a high throughput rate of 628.49 samples per second for the CICIoV2024 dataset and 1039.17 for the N-BaIoT dataset, thus showing its efficiency in detecting attack samples. This efficiency is further supported by the inference latency per sample, where the proposed IDS has an inference latency of only 0.05 ms (ms) for the CICIoV2024 dataset and only 0.043 for the N-BaIoT dataset. The proposed TAPLIDS has a very lightweight architecture, as shown in Table 4. It has a very low model size of only 11.88 KB and 52.00 KB, respectively. Consequently, this analysis proves that the proposed TAPLIDS can be easily deployed in resource-constrained environments without putting any extra burden on their devices.

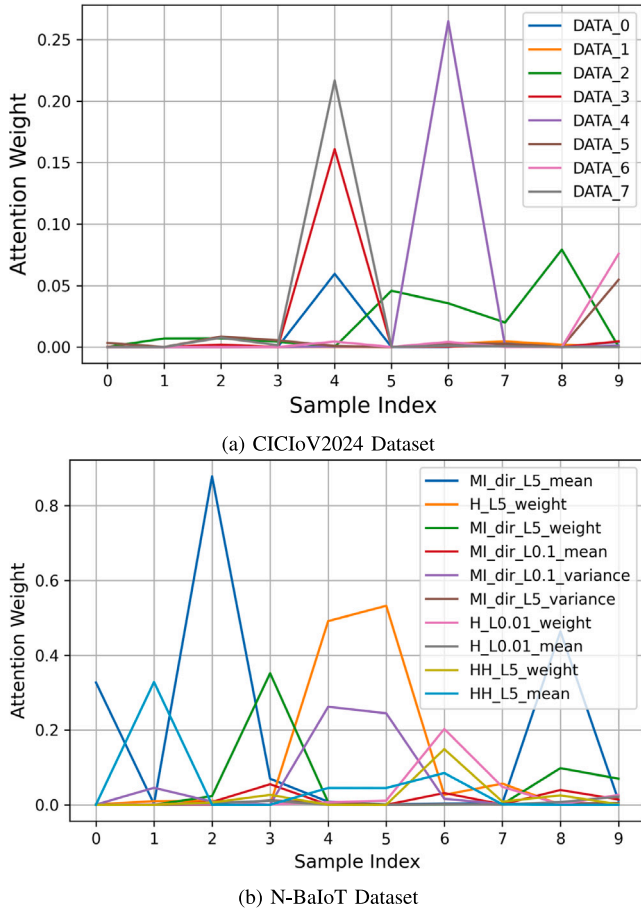


Fig. 9. Feature-level attention weights for 10 test samples.

5.6. Interpretability analysis

In this section, we discuss the interpretability of the proposed TAPLIDS. We first provide the prototype-based distance analysis, followed by the attention visualization. We then discuss the post-hoc interpretability.

5.6.1. Prototype distance visualization

Fig. 8 provides the distance to the closest prototype per class. It presents the minimum Euclidean distance between each test sample's latent representation and the learned class-wise prototypes in the embedding space. For each sample, distances are computed between its latent vector and multiple prototypes per class, and the minimum distance is retained to represent the closest prototype per class. The smaller values indicate higher similarity to the corresponding class prototype in the latent feature space. The variation in distances across classes demonstrates the discriminative structure learned by the model, where samples tend to exhibit lower distances to their true class prototypes compared to others. Overall, the observed separation in distance distributions indicates that the prototype-based latent space effectively captures class-specific representations, enabling reliable class assignment based on nearest-prototype distance minimization.

5.6.2. Attention visualization

We provide the feature importance derived from the learned attention weights of TAPLIDS in Fig. 9. The attention mechanism assigns a weight to each input feature, indicating its relative contribution to the TAPLIDS's decision for each test sample. The attention weights are averaged across all test samples to obtain a global interpretation. We select the top 10 features with the highest mean importance.

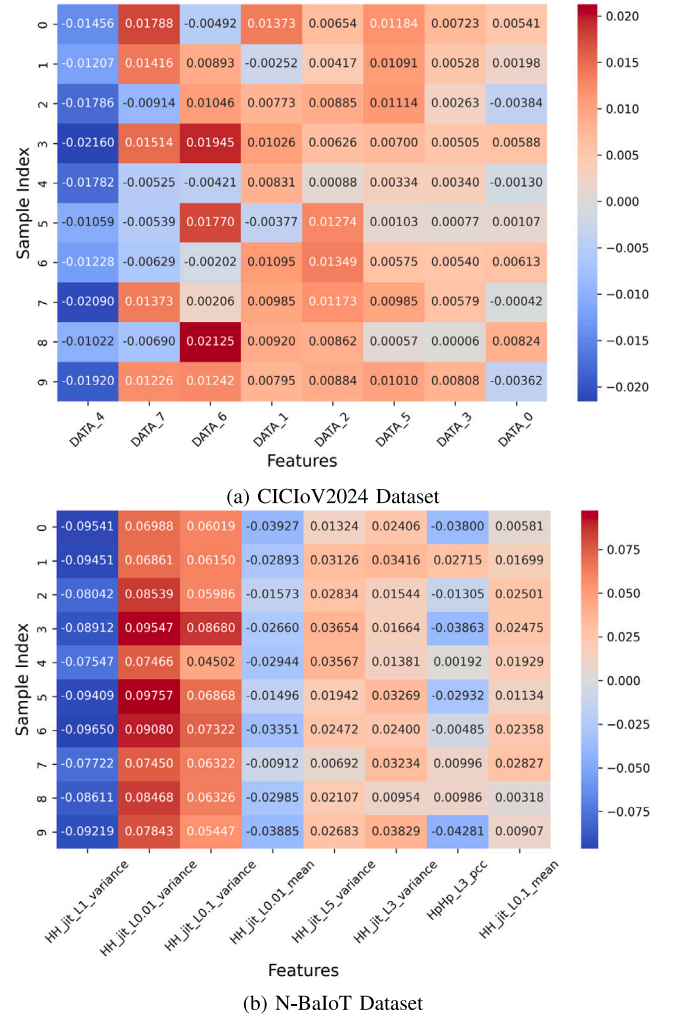


Fig. 10. Top feature contributions to model's decision.

The curves in Fig. 9 show the variation of attention weights for the features across ten random test samples. The higher values indicate a higher contribution of the feature to TAPLIDS's prediction. On the other hand, lower attention values reflect reduced influence. The observed distribution in Fig. 9 highlights that TAPLIDS focuses selectively on a subset of discriminative features rather than treating all inputs equally. This analysis demonstrates that the attention mechanism in TAPLIDS enhances interpretability by enabling feature-level attribution in the learned representation space, which is particularly useful for understanding intrusion patterns in network traffic data.

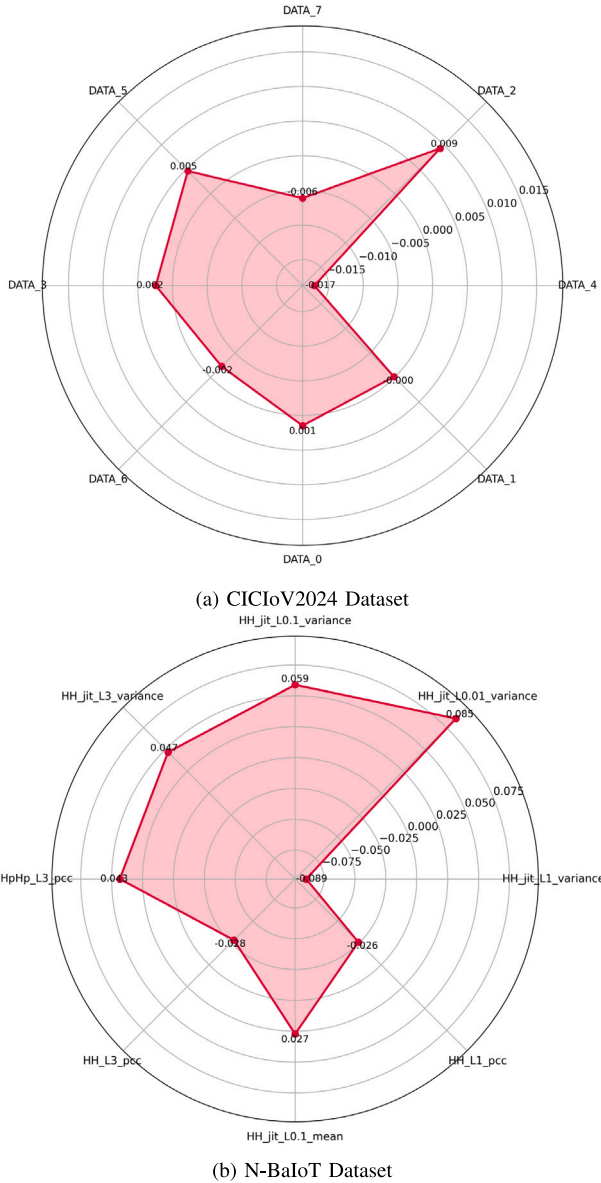
5.6.3. Post-hoc interpretation

In this part, we discuss the post-hoc interpretability. Specifically, we discuss the contribution of features to the TAPLIDS's decision. Fig. 10 provides the heatmap of the top ten highly contributing features across ten samples. Each cell in the figure represents the local importance of a feature for an individual prediction, as estimated by LIME through perturbation-based local surrogate modeling. The positive and higher magnitude values indicate a stronger contribution toward the predicted class, which is shown in the figures as orange. On the other hand, lower or near-zero values suggest a limited influence, as shown in the figure in blue. Features, such as DATA_1, DATA_2, DATA_5, and DATA_3 of the CICIoV2024 dataset in Fig. 10(a) have the highest positive contributions to the TAPLIDS's decision. However, DATA_4 has a negative contribution, while DATA_7 and DATA_6 have positive contributions for some samples and negative contributions for

Table 5

Class-wise recall comparison of TAPLIDS against baselines on CICIoV2024 dataset.

Model	Benign	DoS	Spoofing-RPM	Spoofing-Speed	Spoofing-Steering-Wheel	Spoofing-Gas
Attention-LSTM	0.7710	0.9999	0.7298	0.7991	0.9999	0.9999
Attention-GRU	0.8150	0.9991	0.7300	0.7990	0.9990	0.9999
Attention-BiLSTM	0.7580	0.9999	0.7319	0.7990	0.9999	0.9993
Attention-BiGRU	0.8520	0.9993	0.7328	0.7983	0.9990	0.9997
Proposed TAPLIDS	0.9963	0.9140	0.9569	0.7673	0.9832	0.9932

**Fig. 11.** Local explanation using radar chart.

the others. Regarding the N-BaIoT dataset in Fig. 10(b), features like HH_jit_L0.01_variance and L0.01_variance have the most positive contribution across all test samples, while the rest of the features also show a positive contribution. Only two features, i.e., HH_jit_L1_variance and HH_jit_L0.01_mean, have a negative contribution to TAPLIDS's decision. The variation of feature contributions across samples demonstrates that TAPLIDS relies on different feature subsets for different instances. It highlights TAPLIDS' sensitivity to the local decision boundary and its input-dependent behavior.

Furthermore, Fig. 11 presents the local explanation of TAPLIDS's prediction for a single test instance using the radar plot of LIME. Each

of the axes corresponds to one of the most influential features, and the radial magnitude represents the contribution of each feature to the predicted class in the local surrogate model. Fig. 11 highlights that some features significantly contributed to the prediction, while the remaining features have comparatively lower influence. This indicates that the TAPLIDS's decision for the given instance is driven by a sparse and highly discriminative feature subset rather than uniformly distributed feature importance. The variation in feature contributions in both Figs. 11(a) and 11(b) further reflect the nonlinear decision behavior of the TAPLIDS, where feature importance is highly instance-dependent. Such localized interpretability helps in understanding the specific reasoning behind individual predictions, which is particularly important in IDS.

Overall, Figs. 10 and 11 demonstrate that TAPLIDS consistently focuses on a compact set of discriminative features across samples while dynamically adjusting feature importance at the instance level. Consequently, it shows the strong and robust decision-making capability of TAPLIDS in complex intrusion detection scenarios.

5.6.4. Illustrative explainability case study

To illustrate the explainability mechanisms of TAPLIDS, consider a representative traffic instance classified as a malicious attack. The prototype-based classifier provides class-level evidence through the distances between the instance's latent representation and the learned prototypes, where a smaller distance to prototypes of the predicted class indicates greater similarity to the corresponding learned attack representation. The attention mechanism provides representation-level insight by assigning normalized weights to the dimensions of the dense embedding and using these weights to emphasize selected latent dimensions before the residual representation is formed. LIME complements these mechanisms by providing a local feature-level explanation, identifying input features that positively or negatively influence the individual prediction. Together, these mechanisms provide complementary perspectives on the decision process: prototype distances indicate similarity to learned class representations, attention weights indicate the emphasized dimensions of the learned representation, and LIME relates the prediction to influential input features. This combined interpretation enables security analysts to examine both the learned representation and observable traffic characteristics supporting a predicted attack class.

5.7. Comparison against baselines

We compare the performance of the proposed TAPLIDS against baselines, i.e., BiLSTM, BiGRU, attention-based BiLSTM, and attention-based BiGRU. These baselines have been widely used by researchers to design an efficient intrusion detection framework. We provide the comparison in terms of detection performance and computational complexity. A thorough comparison is as follows.

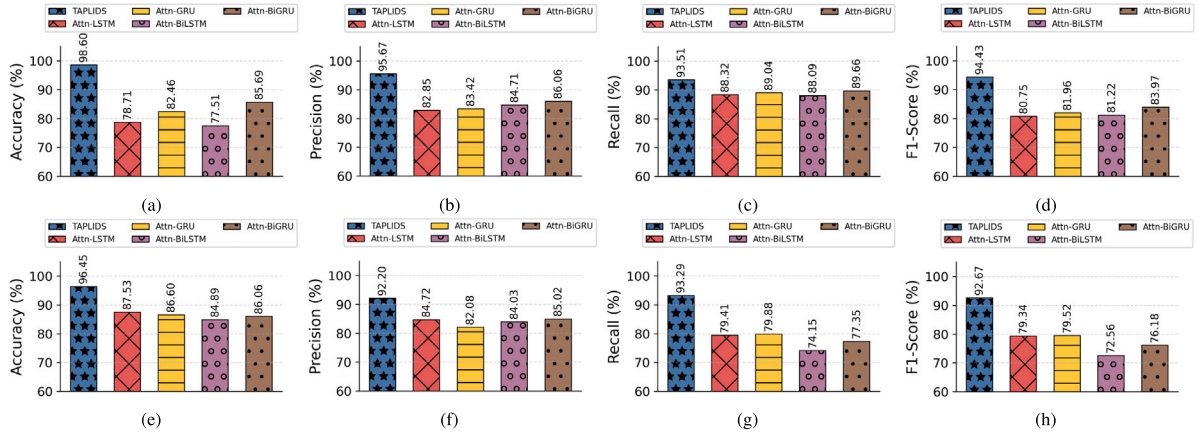
5.7.1. Detection performance comparison

We first provide the class-wise recall (detection rate) comparison against baselines in Tables 5, 6. Table 5 provides the comparison in the CICIoV2024 dataset. The proposed TAPLIDS has achieved higher recall values for all the classes for this dataset, ranging from 0.7673 to 0.9963. On the other hand, the baselines have also performed well, but they achieved low values for some classes of this dataset, i.e., benign,

Table 6

Class-wise recall comparison of TAPLIDS against baselines on N-BaIoT dataset.

Model	Benign	Mirai_udp	Mirai_syn	Mirai_ack	Mirai_scan	Gafgyt_combo	Gafgyt_junk	Gafgyt_scan
Attention-LSTM	0.9990	0.6861	0.9993	0.8750	0.9972	0.7690	0.2119	0.8163
Attention-GRU	0.9981	0.6045	0.9993	0.9622	0.9971	0.6900	0.3650	0.7736
Attention-BiLSTM	0.9991	0.9997	0.9990	0.0043	0.9980	0.7492	0.3614	0.8217
Attention-BiGRU	0.9998	0.9990	0.9991	0.1011	0.9952	0.6090	0.4872	0.9960
Proposed TAPLIDS	0.9988	0.9406	0.9958	0.9701	0.9837	0.8222	0.7795	0.9725

**Fig. 12.** Average performance comparison of the proposed TAPLIDS against baseline frameworks. (Figs. 12(a)–12(d) provide a comparison on the CICIoV2024 dataset, while Figs. 12(e)–12(h) provides a comparison on the N-BaIoT Dataset).

Spoofing-RPM, and Spoofing-Speed. Regarding the N-BaIoT dataset, we provide the comparison in Table 6. The proposed TAPLIDS once again showed efficient performance for all the classes of this dataset, with a highest precision of 0.9988 for the benign class and a lowest of 0.7795 for the Gafgyt_junk class. The baselines, on the other hand, exhibit comparatively weak performance across several attack classes. In particular, Attention-LSTM and Attention-GRU show lower detection performance for the Mirai_udp, Mirai_ack, Gafgyt_combo, and Gafgyt_junk classes. Similarly, the Attention-BiLSTM and Attention-BiGRU models struggle with Mirai_ack, Gafgyt_combo, and Gafgyt_junk classes.

The comparison in both Tables 5 and 6 indicates inconsistent learning of the baseline models across this more complex and imbalanced dataset. In contrast, the proposed TAPLIDS achieves consistently superior performance across all attack classes, demonstrating its robustness and effectiveness in handling imbalanced and diverse intrusion scenarios.

Furthermore, we provide the average comparison of the proposed TAPLIDS against baselines in terms of accuracy, precision, recall, and F1-score in Fig. 12. Figs. 12(a)–12(d) provide the comparison on the CICIoV2024 dataset. The proposed TAPLIDS has outperformed all the baselines on this dataset and achieved an efficient accuracy of 98.60% with a precision of 95.67%, a recall of 93.51%, and an F1-score of 94.43%, respectively. The Attn-BiGRU performed better than the rest of the baselines. The proposed TAPLIDS has comparatively achieved 12.9% higher detection accuracy with a 9.6% higher precision, 3.85% higher recall, and 10.4% higher F1-score compared to the nearest baseline (Attn-BiGRU). Moreover, the comparison on the N-BaIoT dataset is provided in Figs. 12(e)–12(h). The proposed TAPLIDS once again showed an extraordinary performance and achieved a detection accuracy of 96.45%, precision of 92.202.67%, recall of 93.292.67%, and F1-score of 92.67%. The proposed TAPLIDS outperformed the baselines by almost 8.9% higher detection accuracy, 7.4% higher precision, 13.8% higher recall, and 13.3% higher F1-score than the nearest baseline (Attn-LSTM). The Attn-GRU has almost the same performance as Attn-LSTM. The higher values of the proposed TAPLIDS in terms of all the metrics under both datasets further prove its efficiency in detecting threats with high detection accuracy and rate, thereby confirming its suitability for threat detection in ITS environments.

5.7.2. Complexity comparison

Finally, we compare the computational efficiency of the proposed TAPLIDS against baselines in this section. The comparison is conducted using the number of parameters and model size. Fig. 13 provides the complete comparison. The proposed TAPLIDS model achieves a highly compact architecture on the CICIoV2024 dataset with only 3040 parameters and a model size of 11.88 MB, as shown in Fig. 13(a), which is significantly lower than all baseline methods. In contrast, the baseline models contain between 7110 and 23590 parameters, with model sizes ranging from 27.77 KB to 92.15 KB, indicating substantially higher computational and storage overhead. Furthermore, we provide the comparison on the N-BaIoT dataset in Fig. 13(b), which further validates the lightweight nature of the proposed TAPLIDS. TAPLIDS again demonstrates the lowest complexity, with only 13312 parameters and a compact model size of 52.00 KB, significantly outperforming all baseline models. In comparison, the baseline methods require substantially higher resources, with parameters ranging from 17400 to 51016 and model sizes between 67.97 KB and 199.28 KB.

Overall, TAPLIDS consistently achieves a superior trade-off between compactness and model complexity, making it robust for resource-constrained deployments.

5.8. Comparison against modern IDS architectures

In this section, we further evaluate the performance of the proposed TAPLIDS against a transformer-based IDS and a stacking-based ensemble IDS. The Transformer IDS employs an encoder with learnable positional embeddings and multi-head self-attention to model temporal dependencies and feature interactions in network traffic, followed by global pooling and a dense classification layer. The Stacking-based ensemble IDS integrates heterogeneous learners, including Random Forest, XGBoost, and MLP, where probabilistic outputs from base models are combined and passed to a Logistic Regression meta-learner for final prediction. These two baselines represent complementary deep sequence modeling and classical ensemble learning paradigms, enabling a comprehensive evaluation of TAPLIDS's effectiveness.

Table 7 shows the complete comparison of TAPLIDS against these models. The proposed TAPLIDS consistently outperforms both the Transformer-based and stacking-based ensemble IDS across both

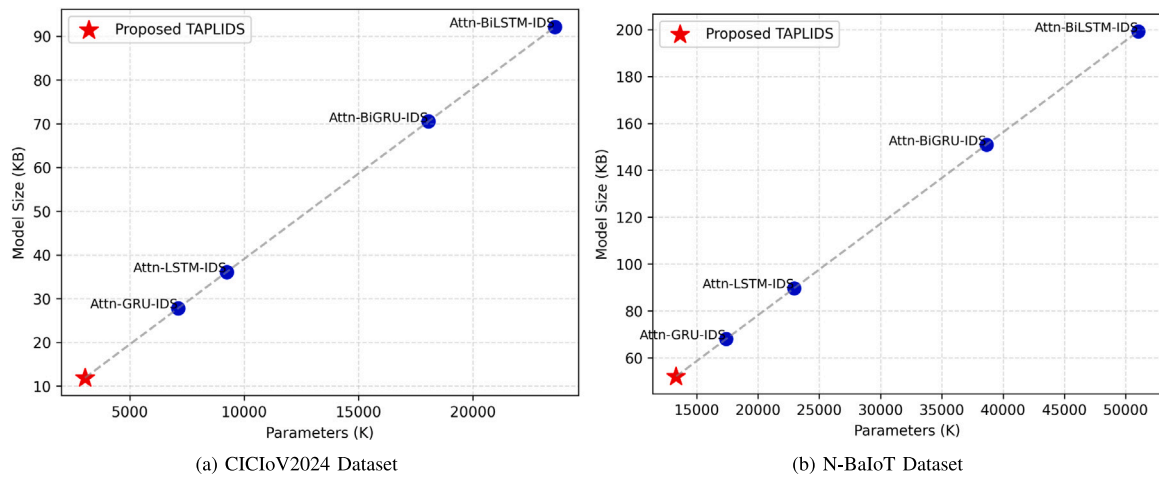


Fig. 13. Comparison against baselines regarding model complexity.

Table 7

Comparison against modern IDS architectures.

Method	Accuracy	F1-score	Model Size	Inference latency	Interpretability	Deployment Feasibility
CICIoV2024 Dataset						
Transformer-based IDS	95.61%	92.53%	0.72 MB	0.069 ms/sample	×	Medium-High
Stacking-based Ensemble IDS	97.66%	93.53%	5.84 MB	0.002 ms/sample	×	Low-Medium
Proposed TAPLIDS	98.59%	94.43%	11.88 KB	0.050 ms/sample	✓	High-Very High
N-BaIoT Dataset						
Transformer IDS	86.95%	84.69%	0.80 MB	0.141 ms/sample	×	Medium-High
Stacking-based Ensemble IDS	91.42%	90.69%	13.17 MB	0.003 ms/sample	×	Low-Medium
Proposed TAPLIDS	96.45%	92.67%	52.00 KB	0.043 ms/sample	✓	High-Very High

datasets. Specifically, TAPLIDS achieves an accuracy of 98.60% and an F1-score of 94.43% on the CICIoV2024 dataset, representing improvements of 2.99% and 1.90% over the Transformer-based IDS and 0.94% and 0.90% over the stacking-based ensemble IDS. Similarly, on the N-BaIoT dataset, TAPLIDS attains 96.45% accuracy and a 92.67% F1-score, outperforming the Transformer baseline by 9.50% and 7.98%, and the ensemble baseline by 5.03% and 1.98%, respectively. In addition to its superior detection performance, TAPLIDS demonstrates remarkable computational efficiency. It requires only 11.88 KB on the CICIoV2024 dataset and 52.00 KB on the N-BaIoT dataset, compared with 0.72–0.80 MB for the Transformer model and 5.84–13.17 MB for the stacking-based ensemble. Although the ensemble model exhibits the lowest inference latency (0.002–0.003 ms/sample), TAPLIDS maintains a competitive latency of 0.043–0.050 ms/sample while providing substantially higher detection performance, significantly lower storage requirements, and intrinsic interpretability. These results indicate that TAPLIDS achieves a more favorable trade-off between detection performance, computational efficiency, and practical deployability, making it particularly suitable for resource-constrained ITS environments.

5.9. Sensitivity analysis of TAPLIDS

In this section, we present the sensitivity analysis of the proposed TAPLIDS framework. Specifically, we first evaluate the impact of different temporal window sizes and then analyze the sensitivity of the framework to different prototype configurations.

5.9.1. Sensitivity analysis of temporal window size

We evaluated the sensitivity of TAPLIDS to different temporal configurations by conducting experiments on the CICIoV2024 dataset with different temporal window sizes while keeping all other experimental settings unchanged. Specifically, window sizes of 3, 5, 7, 10, and 12 were evaluated with a fixed stride of one. The performance of each

Table 8

Sensitivity analysis of TAPLIDS to different temporal window sizes.

Window Size	Accuracy	Macro Precision	Macro Recall	Macro F1-score
3	98.60%	95.67%	93.51%	94.43%
5	98.49%	95.21%	92.40%	93.76%
7	97.84%	93.21%	91.48%	92.32%
10	98.03%	94.06%	91.46%	92.65%
12	98.47%	94.25%	93.25%	93.68%

configuration was assessed using accuracy, macro-precision, macro-recall, and macro F1-score. The results are reported in Table 8 and provide an empirical assessment of the effect of temporal context on the detection performance of TAPLIDS. The results show that the selected window size of $W = 3$ achieves the best overall performance, with 98.60% accuracy and a 94.43% macro F1-score. Increasing the window size to 5, 7, 10, and 12 results in lower macro F1-scores of 93.76%, 92.32%, 92.65%, and 93.68%, respectively. This indicates that extending the temporal context beyond three observations does not provide additional detection benefits for the evaluated data. Larger windows may aggregate observations over broader temporal intervals and consequently dilute short-duration variations associated with attack behavior. Therefore, $W = 3$ provides an effective temporal context for capturing relevant short-term variations while maintaining strong detection performance.

5.9.2. Prototype number sensitivity analysis

To investigate the sensitivity of the proposed multi-prototype classification strategy to the number of prototypes assigned to each class, we evaluated TAPLIDS using different prototype configurations. Specifically, the number of prototypes per class was varied as $K \in \{1, 2, 3, 4, 5\}$, while all other architectural and experimental settings were kept unchanged. Here, K denotes the number of learnable prototypes maintained for each class, resulting in a total of $C \times K$ prototypes for a

Table 9

Sensitivity of TAPLIDS to the number of prototypes per class.

K	Accuracy	Macro Precision	Macro Recall	Macro-F1-score
1	97.91%	94.05%	91.95%	92.77%
2	97.93%	92.91%	91.53%	92.188%
3	98.60%	95.67%	93.51%	94.43%
4	97.97%	92.48%	92.98%	92.64%
5	97.34%	90.95%	91.03%	90.95%

Table 10

Ablation study of TAPLIDS.

Configuration	Accuracy	Macro Precision	Macro Recall	Macro F1-score
Full Model	98.60%	95.67%	93.51%	94.43%
w/o Attention	97.17%	90.28%	90.84%	90.49%
w/o Residual	97.73%	92.62%	92.00%	92.12%
w/o Attention + Residual	96.32%	91.08%	85.14%	87.70%

classification problem with C classes. The evaluation was performed using Accuracy, Macro Precision, Macro Recall, and Macro-F1 to assess the effect of prototype capacity on overall and class-balanced classification performance. Table 9 demonstrates the output regarding varying K values. The number of prototypes has a noticeable effect on the classification performance. Using a single prototype per class ($K = 1$) achieves 97.91% accuracy and 92.77% Macro-F1, while increasing the number to $K = 2$ provides only a marginal change in performance. The best performance is obtained with $K = 3$, achieving 98.60% accuracy, 95.67% Macro Precision, 93.51% Macro Recall, and 94.43% Macro-F1. This suggests that three prototypes provide an appropriate representational capacity for the evaluated traffic patterns. Increasing the number of prototypes beyond $K = 3$ does not provide further improvement. In particular, $K = 4$ reduces the Macro-F1 to 92.64%, while $K = 5$ results in a further decrease to 90.95%. These results suggest that excessive prototype allocation may introduce redundant or less discriminative representations rather than providing additional performance benefits. Therefore, $K = 3$ is selected as the default configuration for TAPLIDS, providing the best overall classification performance among the evaluated prototype configurations.

5.10. Ablation study

To assess the contribution of the attention and residual components, an ablation study was conducted on the CICIoV2024 dataset by selectively removing each component from TAPLIDS. As reported in Table 10, the full model achieves the best overall performance, obtaining 98.60% accuracy and 94.43% macro-F1. Removing the attention mechanism decreases the macro-F1 to 90.49% and accuracy to 97.17%, demonstrating that attention-based feature weighting is important for identifying discriminative features. Removing the residual connection results in 97.73% accuracy and 92.12% macro-F1, indicating that the residual pathway also contributes to the learned representation, although its individual impact is smaller than that of attention. The simultaneous removal of both components causes the largest performance degradation, reducing macro-recall from 93.51% to 85.14% and macro-F1 from 94.43% to 87.70%. Overall, these results confirm that both components contribute to the effectiveness of TAPLIDS, with the attention mechanism providing the stronger individual contribution.

6. Conclusion

This paper proposed TAPLIDS, a temporal-aware prototype-based lightweight and interpretable intrusion detection framework for intelligent transportation systems. TAPLIDS addresses key limitations of existing IDS by capturing temporal dependencies through sliding-window feature extraction, handling heterogeneous data via attention-based embedding, preserving information with residual connections, and modeling diverse attack behaviors using multi-prototype classification. Extensive experiments on CICIoV2024 and N-BaIoT datasets

demonstrate that TAPLIDS achieves superior detection performance (98.60% and 96.45% accuracy, respectively) with exceptionally low computational complexity (as few as 3040 parameters and 11.88 KB model size). Furthermore, attention weights, prototype distances, and LIME explanations provide multi-level interpretability, enabling trusted deployment in safety-critical ITS environments. A limitation of the current study is that long-term adaptation to evolving and heterogeneous ITS environments is not explicitly investigated. Future work will therefore consider adversarial robustness, concept drift, continual learning, imbalanced and emerging attacks, and long-term model updating and maintenance. Federated learning will also be explored for privacy-preserving distributed security, together with graph-based approaches and large-scale real-world deployment.

CRedit authorship contribution statement

Danish Javeed: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Muhammad Adil:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization, Visualization, Data curation. **Shifa Shoukat:** Writing – review & editing, Conceptualization, Visualization, Data curation. **Muhammad Shahid Saeed:** Writing – review & editing, Validation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Danish Javeed (Member, IEEE) received the Ph.D. degree in Software Engineering from the Software College, Northeastern University, China, in 2024, and the M.E. degree in Computer Applied Technology from Changchun University of Science and Technology, China, in 2020, both under prestigious fellowships funded by the Ministry of Education, Government of China. He is currently a Postdoctoral Researcher with the College of Artificial Intelligence, Dalian Maritime University, China. His research interests focus on AI-driven cybersecurity, industrial and IoT security, intrusion detection and prevention, LLM-enabled human-in-the-loop defense, deep and federated learning, and explainable AI for trustworthy cyber-defense. He has authored or co-authored 40+ papers in leading journals and conferences, including IEEE Communications Surveys & Tutorials, IEEE Trans. on Intelligent Transportation Systems, IEEE Trans. on Network Science and Engineering, IEEE Trans. on Consumer Electronics, IEEE Internet of Things Journal, IEEE Computer, ACM Trans. on Internet Technology, Computers & Security, Computer Networks, Decision Support Systems, and IEEE ICC.



Muhammad Adil received the M.E. degree in computer science and technology from Dalian University of Technology (DUT), China, in 2022. He is currently pursuing a Ph.D. degree in computer science at the School of Computer Science and Technology, Tianjin University, China. His research interests include time-sensitive industrial networks, 5G, federated learning, and cybersecurity. He has received several awards, including the Chinese Government Scholarship from the Ministry of Education. He has authored or co-authored 20+ papers in high-impact journals and conferences, including IEEE COMST, IEEE T-ITS, IEEE TNSE, IEEE IoTJ, IEEE Computer, ACM ToIT, COSE, FGCS, Adhoc Networks, Cluster Computing, and IEEE ICCT.



Shifa Shoukat is currently pursuing a Ph.D. degree in Software Engineering with the Software College, Northeastern University, China, under the prestigious fellowship of the Ministry of Education funded by the Government of China. She got her M.S. degree from Quaid-i-Azam University, Islamabad, Pakistan, in 2022. Her research interests include Information Security, Intrusion Detection and Prevention systems, the Internet of Things, Deep Learning, Explainable AI, and Software-defined Networking. She has authored or co-authored 10+ papers in high-impact journals and conferences, including IEEE IoTJ, COSE, IEEE ICC, IEEE ICCT, and Cluster Computing.



Muhammad Shahid Saeed is currently pursuing a Ph.D. degree in Software Engineering with Dalian University of Technology, Dalian, China, under the prestigious fellowship of the Ministry of Education funded by the Government of China. His research interests include Information Security, Intrusion Detection and Prevention systems, the Internet of Things, and Deep Learning. He has authored or co-authored 10+ papers in high-impact journals and conferences, including IEEE IoTJ, IEEE TCE, ACM ToIT, COSE, and FGCS.